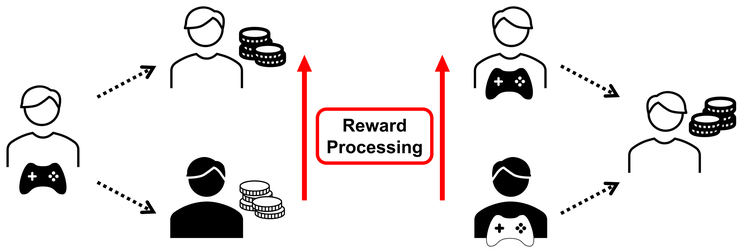
Interactions between Agency and Earning Money in Reward Learning

Mathew Rocha Hammerstrom[1], Maren Giersiepen[2], & Olave E. Krigolson[1]

Theoretical and Applied Neuroscience Laboratory, University of Victoria[1]; Body Social Cognition and Action Lab, LMU Munich[2]

INTRODUCTION

Humans preferentially process rewards for themselves compared to others but also want control over learning.



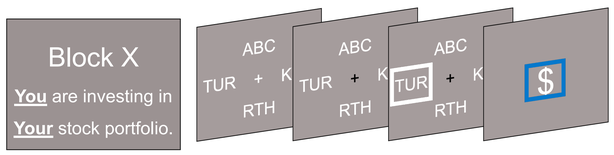
Which is more important for reward learning: agency or earning rewards?

METHODS

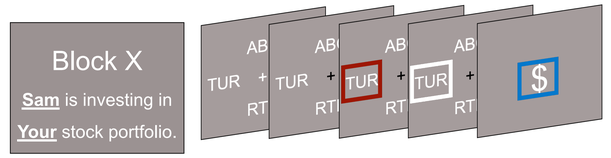
Player must learn which option is most likely to yield rewards.

24 Blocks
20 Trials
$n_{current}$ == 21

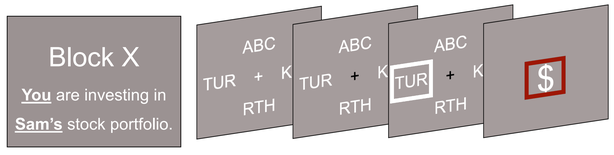
Standard
Participant plays and earns money



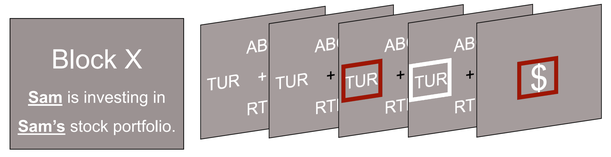
Nervous
Someone else plays and the participant earns money



Charity
Participant plays and someone else earns money



Observational
Someone else plays and earns money

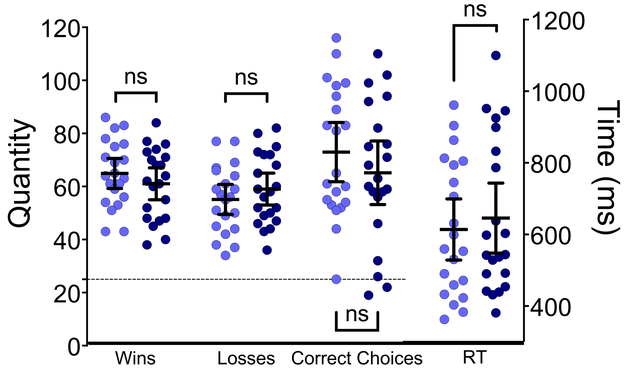


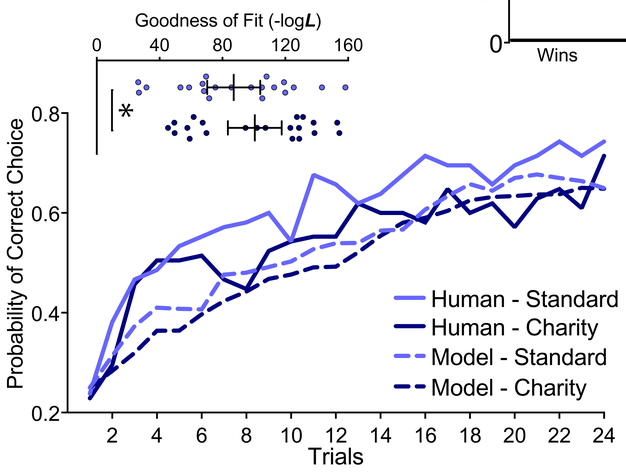
1) Modelling behaviour measures learning.

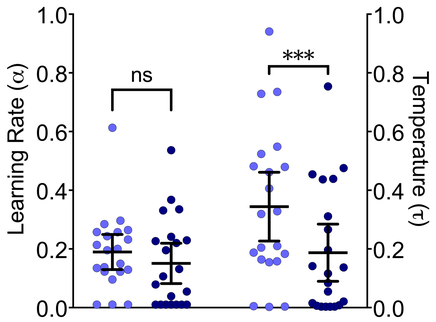
2) ERPs linked to feedback measure reward processing.

2 (**Agency**) x 2(**Earning**)

Participants display overall similar behaviour, but modelling suggests they make more exploratory decisions with other's money.

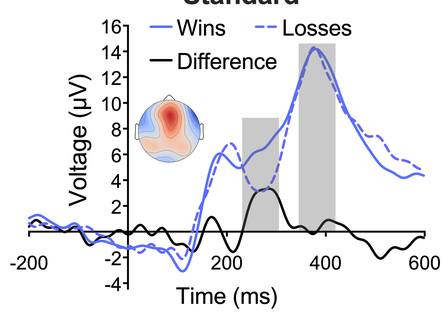




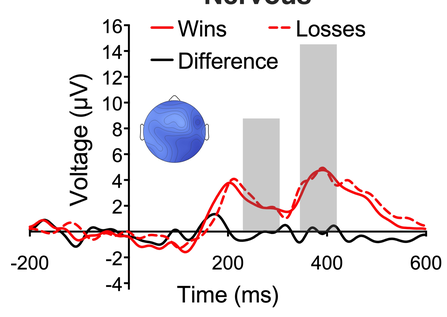


RESULTS

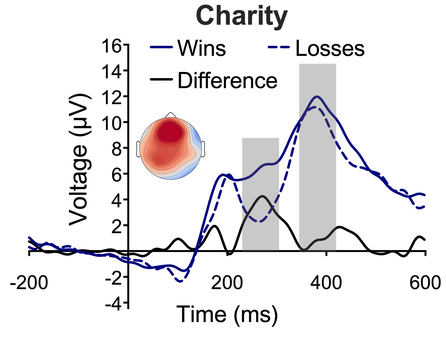
Standard



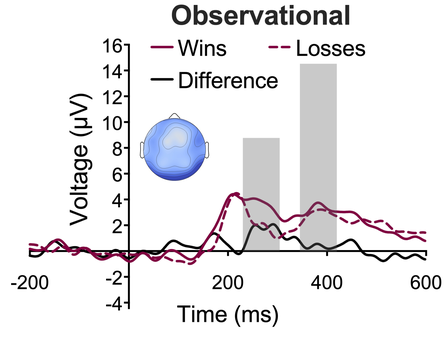
Nervous



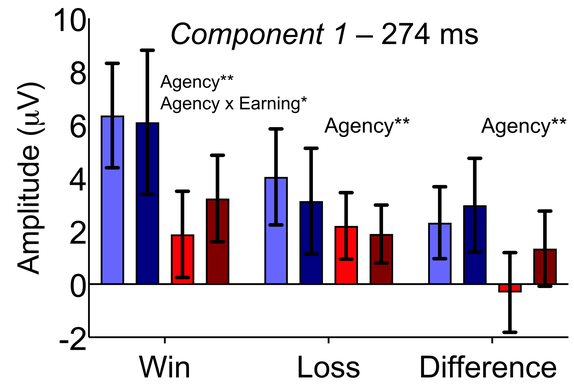
Charity



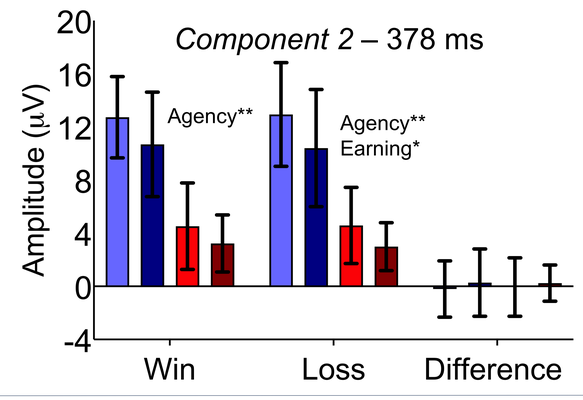
Observational



Component 1 – 274 ms



Component 2 – 378 ms



CONCLUSIONS

1. Participants make more exploratory decisions with other's money but behave relatively similar overall.
2. The **agency** bias ("I decide" > "They decide") is larger than the **earning money** bias ("I win" > "They win") in reward processing.
3. Results are limited by task difficulty and non-learners.